

S&DS 265 / 565

Introductory Machine Learning

Language Models

February 23

Yale

Business items

- Assignment 3 is out
- Quiz 3 on Wednesday

Midterm review sessions

Ruiqi: Friday, Feb 27, 7:00pm, KT 205

Jianliang: Saturday, Feb 28, 5:00pm, KT 1105B

Dingyao: Monday, March 2, 8:00pm, KT 205

John: Tuesday, March 3, 5:30pm, KT 205

Invitation: Panel discussion

Class on Monday, April 20:

We will have a panel discussion on

Societal issues for AI and Machine Learning

If you would like to volunteer to participate, please email us:

To: `sds265@yale.edu`

Subject: `iML panel`

For Today

- Language models
- Concepts...no new methods

Language models

- A language model is a way of *generating* any sequence of words

$$\begin{aligned} P(\text{"the whole forest had been anesthetized"}) = & \\ & P(\text{"the"}) \times P(\text{"whole"} \mid \text{"the"}) \\ & \times P(\text{"forest"} \mid \text{"the whole"}) \\ & \times P(\text{"had"} \mid \text{"the whole forest"}) \\ & \times P(\text{"been"} \mid \text{"the whole forest had"}) \\ & \times P(\text{"anesthetized"} \mid \text{"the whole forest had been"}) \end{aligned}$$

Remixing Noon

Text generated from Channel Skin by Jeff Noon

"The whole forest had been anesthetised, her temples wired, her senses stimulated, her eyes were not on Eva, not on Eva, not on Eva, not on anybody in that same realm, the land of dreams and nightmares."

Viability: 0.000000326%



<https://chatbotslife.com/notes-on-remixing-noon-generative-text-and-markov-chains-84ff4ec23937>

Text generation

- Words generated one-by-one
- A word is chosen by sampling from a probability distribution
- Result is purely synthetic text—generative AI

Uses of language models

- Speech recognition
- Machine translation
- Text compression
- Texting
- Email completion
- Image captioning
- Mind reading from fMRI

Dasher: LMs for assistive devices

Language models enable new modes of text input:

https://www.youtube.com/watch?v=quw_Kci4fUg

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- The number of *histories* grows as V^{n-1} . Number of parameters in model grows as V^n , where V is number of words in vocabulary.
- What are some ways of reducing the number of parameters?

One approach: Grouping histories

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- What are some example groupings?

Grouping histories

- Unigrams: $g(w_1, \dots, w_n) = \emptyset$.

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- Trigrams: $g(w_1, \dots, w_n) = (w_{n-1}, w_n)$.
- Number of parameters grows as $O(V)$, $O(V^2)$, and $O(V^3)$, respectively.

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Estimating parameters

- The maximum likelihood estimate of a trigram model:

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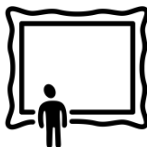
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- $\text{count}(w_1, w_2, w_3)$ is just the number of times the trigram appears in a large corpus of text.
- What are some problems with this model?

Sparse data problem



The next group of slides presents one way of quantifying the problem of sparse data in language modeling

Half Earth

Aug 06, 2018 by Foundation Staff 0 Comment Google Earth, Half-Earth Project, Map of Life

By Jeremy Malczyk, Michelle Duong, Ajay Ranipeta, Chris Heltne, Walter Jetz of Map of Life, Yale University, and the E.O. Wilson Biodiversity Foundation Half-Earth Project

This article originally in *Medium*, July 30, 2018



Learn how you can be part of bringing Half-Earth to life.

The Significance of Biodiversity



Narrow-billed Tody, *Todus angustirostris*. Photo by Julie Hart.

<https://eowilsonfoundation.org/mapping-species-for-half-earth/>

<https://www.half-earthproject.org/>

Specious species



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- What if Wilson observes 100 unique species?

Missing species: Good-Turing

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- Good-Turing: Estimate of the probability that the next animal is a new species? $\hat{p}_{GT} = 100/100,000 = 10^{-3}$.
- This is an estimate of the missing probability mass.

Good-Turing and leave-one-out

- The Good-Turing estimate is

$$\hat{p}_0 = \frac{N_1}{N}$$

where N_1 is the number of species observed exactly once, and N is the total number of observations.

- This can be thought of (and justified) using leave-one-out.
- When we “leave out” each of the N_1 unique observations, and then predict, it is previously unseen

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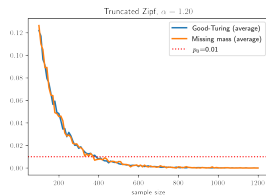
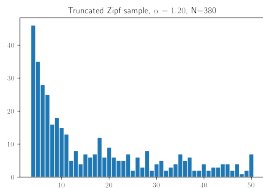
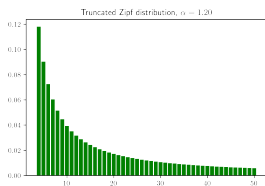
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- When I see a new trigram, 10% of the time it won't have been seen before. (This is a typical number.)
- This means that the maximum likelihood estimate (MLE) is zero, and the probability that my model predicts the next word will be zero.
- The MLE is supported on the observed data. We need to spread out the probability over unseen events.

Good-Turing is remarkably accurate



Good-Turing estimate of the missing mass for a heavy-tailed distribution

Estimating parameters

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- How can the model be strengthened?

Smoothing and Interpolation

Smoothing:

$$\tilde{p}(w_3 | w_1, w_2) = \frac{\text{count}(w_1, w_2, w_3) + \varepsilon}{\text{count}(w_1, w_2) + V\varepsilon}$$

Smoothing and Interpolation

Linear interpolation:

$$p(w_3 | w_1, w_2) = \lambda_3 \hat{p}(w_3 | w_1, w_2) + \lambda_2 \hat{p}(w_3 | w_2) + \lambda_1 \hat{p}(w_3)$$

where $\lambda_1 + \lambda_2 + \lambda_3 = 1$.

This is a type of “mixture model”

How good is a language model?

The next group of slides presents a useful way of quantifying how good a language model is

Recall: Geometric mean

The *arithmetic mean* of $1/4$, 4 , 8 is

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$$\sqrt[3]{\frac{1}{4} \cdot 4 \cdot 8} = 2$$

The geometric mean is no greater than the arithmetic mean

Recall: Geometric mean

The *geometric mean* of $x_1, \dots, x_n$ is

$$\sqrt[n]{x_1 x_2 \cdots x_n} = \left(\prod_{i=1}^n x_i \right)^{\frac{1}{n}}$$

How good is a language model? Perplexity

Perplexity is defined as

$$\begin{aligned}\text{Perplexity}(\theta) &= p_{\theta}(w_1 w_2 \cdots w_N)^{-\frac{1}{N}} \\ &= \left(\prod_{i=1}^N p_{\theta}(w_i \mid w_{1:i-1}) \right)^{-\frac{1}{N}}\end{aligned}$$

where $w_1, w_2, \dots, w_N$ is a large chunk of text that wasn't used to train the language model.

How good is a language model? Perplexity

- Perplexity is the inverse of the geometric mean of the word probabilities
- Intuitively, if the perplexity is 20, the actual next word is in the top 20 words of the model (on average).
- This is the (geometric) average “branching factor” for the model on real text

perplexity

Ask anything...



Model ▾



💰 Finance

✈️ Travel

🔧 Troubleshoot

♥️ Health

🔍 Research

Recall: Embeddings

We discussed embeddings as constructed from simple bigram language models:

$$\log \mathbb{P}_{\text{near}}(w_1, w_2) = \varphi(w_1)^T \varphi(w_2) + \text{constant}$$

where $\varphi(w)$ is the *embedding vector* for word w .

Learning embeddings

- This language model is trained using stochastic gradient descent.
- If we have $d = 500$ dimensional embeddings, there are 500 trainable parameters for *each* word
- How are these numbers adjusted (trained)?

Learning embeddings: SGD

Start with *random* embeddings. Then iterate many, many times:

- 1 Sample a pair of words w_1 and w_2 that appear nearby
- 2 Adjust the embedding parameters of each word:

$$\varphi(w_1) \leftarrow \varphi(w_1) + \eta \varphi(w_2)$$

$$\varphi(w_2) \leftarrow \varphi(w_2) + \eta \varphi(w_1)$$

The vectors are brought closer together in each step!

Learning embeddings: SGD

Why is this? The *loss* is the *negative* log-probability:

$$-\log \mathbb{P}_{\text{near}}(\mathbf{w}_1, \mathbf{w}_2) = -\varphi(\mathbf{w}_1)^T \varphi(\mathbf{w}_2) + \text{constant}$$

The derivative of this loss with respect to $\varphi(\mathbf{w}_1)$ is $-\varphi(\mathbf{w}_2)$

SGD update adds a small step size η times *minus* the derivative:

$$\varphi(\mathbf{w}_1) = \varphi(\mathbf{w}_1) + \eta \varphi(\mathbf{w}_2)$$

Modern language models

An LLM assigns a “score” to possible next words:

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The function $s(v; w_{1:n})$ is learned on large amounts of text (unsupervised) using a type of deep neural network called a *transformer*.

Modern language models

In fact, the score is computed as

$$s(v; \underbrace{w_1, \dots, w_n}_{\text{word history}}) = \varphi(v)^T g(w_1, \dots, w_n)$$

- the “grouping function” $g(w_1, \dots, w_n)$ is like an embedding, that maps the word history to a high dimensional vector
- The vector $\varphi(v)$ is an embedding of word v

We'll see how this is done when we discuss neural networks

Summary of today: Language models

- A language model is used to predict or generate the next word
- Used in many different applications
- Probabilities need to be “smoothed” to avoid zeros
- Perplexity is a measure of a language model’s predictive power